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Genomic analysis reveals recent population decline and exceptionally low genome-wide heterozygosity of the critically endangered Philippine eagle, *Pithecophaga jefferyi* (Aves: Accipitridae)

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Abstract

Background The Philippine eagle (*Pithecophaga jefferyi*) is one of the rarest raptors in the world and is the national bird of the Philippines. It is listed as Critically Endangered by the International Union for Conservation of Nature and has been the subject of a captive breeding program, which started more than 30 years ago to ensure the survival and conservation of the species. To estimate the genetic diversity and infer the demographic history of the Philippine eagle, we sequenced and analyzed genomes of 35 individuals under the care of the Philippine Eagle Foundation.

Results Phylogenetic analysis placed *P. jefferyi* within the subfamily Circaetinae of Accipitridae. Demographic history reconstruction revealed two historical bottlenecks and an ongoing population decline, which was found to predate documented deforestation in the Philippines. This suggests that unobserved natural or anthropogenic factors might have led to declines in the Philippine eagle population long before current habitat degradation. Genome-wide heterozygosity was the lowest measured among raptors. This highlights the precarious genetic state of the Philippine eagle, as low heterozygosity raises risks of inbreeding depression and increased vulnerability to diseases and habitat loss.

Conclusions This study demonstrates that genetic diversity of the Philippine eagle is critically low, indicating that breeding and translocation programs must explicitly account for genetic effects to maintain demographic stability and support long-term genetic resilience of this critically endangered species.

Keywords Biodiversity, Demographic history, Endangered, Genetic diversity, Genome, Philippine eagle

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Background

The Philippine eagle (*Pithecophaga jefferyi*) is one of the largest and most powerful forest raptors [1], and is officially recognized as the national bird of the Philippines [2]. It is found across four islands: Mindanao, Leyte, Samar, and Luzon, though its population is now fragmented and scattered across montane and lowland tropical forests [3, 4]. As a forest-dependent raptor needing large land areas for hunting territory, it is susceptible to habitat destruction and fragmentation [5, 6]. Because of this, the Philippine eagle is classified as 'Critically Endangered' (CR) in both the IUCN Red List and the Philippine Red List DAO 2019–09, which makes it one of the most threatened birds globally [7, 8]. Its population has drastically declined over the past five decades, primarily due to habitat loss from deforestation, agricultural expansion, and mining, with many planned mining concessions overlapping spatially with high-suitability nesting areas [4, 9, 10]. Mindanao Island remains the species' stronghold, hosting the largest population [9, 10]. Recent estimates, derived from Species Distribution Models (SDMs) and home-range data, suggest a global population of approximately 392 breeding pairs, or 784 mature individuals [4], emphasizing the urgent need for comprehensive conservation strategies.

Much like other imperiled raptors in Mindanao, the Philippine eagle acts as a sentinel species, whose population status can indicate ecosystem health. Additionally, targeting conservation actions to the raptor could benefit a wide array of co-occurring taxa as an umbrella species [11–13]. Despite its ecological significance, population genetic studies on its molecular information have been limited. Previous studies have reported haplotype and nucleotide diversity of the mitochondrial control region of captive Philippine eagles [14], and more recent studies using partial mitogenomes elucidated genetic diversity and population structure within Mindanao [15].

Studies of the smaller mitochondrial genome, although cost-effective and widely applied, may not fully capture patterns of variation found in the nuclear genome [16–18]. Although a draft genome for the Philippine eagle has previously been deposited in NCBI [19], it lacks quality and provenance, which limits its utility for population genetic analyses. Local initiatives have demonstrated the value of genomic resources for other endemic and endangered Philippine species [20], yet comprehensive population-level analyses utilizing nuclear genomic data remain limited for the Philippines.

To address this gap, we estimated genome-wide heterozygosity and reconstructed the species' demographic history of the Philippine eagle [21] using a newly generated genome assembly. This dataset represents the initial phase of a broader effort to generate resources for the integration of genomic information in conservation

and breeding programs [22], serving as a reference for designing molecular markers that can inform long-term monitoring and population management of the Philippine eagle [7, 11] and support strategies for conservation of the eagle and other co-occurring endemic species [11].

Methods

Sample collection and DNA extraction

Blood samples were collected from thirty-five Philippine eagles at the Philippine Eagle Center of the Philippine Eagle Foundation in Davao City, Philippines (Supplementary Table S1 and Figure S1). Samples were transported under the Wildlife Transport Permit No. DVO-WLTP-202400251 from the Department of Environment and Natural Resources (DENR). The samples were collected in K2 EDTA tubes and stored at 4 °C before DNA extraction using the DNeasy Blood and Tissue Kit (Qiagen, Venlo, The Netherlands). DNA quality and quantity were assessed using agarose gel electrophoresis (AGE, 1.2% agarose), spectrophotometry (NanoDrop, ThermoFisher Scientific, Waltham, USA), and fluorometry (Qubit 2.0 Fluorometer, ThermoFisher Scientific). Extracted DNA was stored at –20 °C until further processing.

Library preparation and sequencing

Genomic libraries were constructed from 100 ng of high-quality genomic DNA using the MGIEasy FS DNA Library Prep Kit (MGI, Shenzhen, China). Library concentrations were quantified using the Qubit dsDNA Broad Range Assay (ThermoFisher Scientific), and average fragment sizes were determined using the TapeStation 2000 (Agilent Technologies, Santa Clara, USA). Libraries were prepared for DNA Nanoball sequencing using the MGIEasy Circularization Module (MGI) and sequenced on the DNBSEQ-G400 platform, generating 150-bp paired-end reads with approximately 30× genome coverage per individual (Supplementary Table S2).

Genome assembly and quality assessment

Raw sequencing reads for each Philippine eagle sample were quality-trimmed using FASTP v1.0.1, and individual genomes were assembled de novo using SOAP denovo2 v2.4.2 with k-mer sizes of 61, 71, 81, and 91 [23, 24]. Assembly quality was evaluated using QUAST v5.2.0, with the 71-mer genome assemblies showing optimal assembly metrics [25]. The assemblies were assessed for their contiguity through the N50 statistic, or the sequence length of the contig at which 50% of the genome assembly is covered; the largest contig size, and the number of ambiguous bases (Ns) per 100,000 base pairs (100kbp) (Supplementary Table S3).

We also evaluated the biological completeness of the de novo assemblies using BUSCO (Benchmarking Universal

Single-Copy Orthologs) v5.7.0 [26]. This tool identifies highly conserved genes, which are genes expected to appear as single copies in any intact genome, and uses them as a proxy for overall assembly quality. The BUSCO aves_odb10 database [26], which contains 8,338 orthologs curated from diverse bird species, was used for assessing the completeness of the assemblies. In assessing completeness and genomic integrity, the proportion of complete (single-copy or duplicated), fragmented, or missing orthologs are calculated (Supplementary Figure S2).

All contig-level assemblies generated by SOAPdenovo2 from the 35 Philippine eagles were compiled into a single FASTA file. This produced extended fragments that approximate the collective sequence data of the sampled population. These fragments were then used as input for the assemblers Flye v2.9.1 and Redbean v2.5.0, which are optimized for continuous, high-length input, resulting in higher-quality meta-assemblies that integrate information across samples [27, 28].

Contiguity of the Flye and Redbean meta-assemblies were again assessed with QUAST v5.2.0, and completeness was assessed using the BUSCO v5.7.0 aves_odb10 database [26]. While Redbean produced superior QUAST metrics, BUSCO determined that Flye produced an assembly with more complete avian orthologs. Consequently, the Flye meta-assembly was selected for downstream analyses based on better gene completeness (Supplementary Table S4).

Genome annotation

Transfer RNAs and ribosomal RNAs were predicted using tRNAscan-SE v2.0.12 and barrnap v0.9, respectively [29, 30]. Repetitive elements were identified and subsequently masked using RepeatMasker v4.1.6 [31]. To exclude sex-linked sequences, the repeat-masked genome was mapped to the golden eagle (*Aquila chrysaetos*) reference assembly GCF_900496995.4 using minimap v2.28, enabling identification of contigs associated with the Z and W chromosomes [32]. Only autosomal contigs were retained for gene annotation to avoid any errors introduced by the sex chromosomes [33].

A gene annotation strategy integrating multiple prediction approaches was employed through EvidenceModeler (EVM) v2.1.0 [34]. Homology-based prediction with GeMoMa v1.9 and protein sequences from three closely related Accipitridae species: golden eagle (GCF_900496995.4), white-tailed eagle (*Haliaeetus albicilla*, GCF_947461875.1), and harpy eagle (*Harpia harpyja*, GCA_026419925.1) [35]. For ab initio prediction, two AUGUSTUS v3.5.0 models were trained and applied: one using raptor sequences (derived from GeMoMa output) and another using chicken (*Gallus gallus*) sequences [36]. The autoAug workflow was utilized,

which encompassed custom training set generation (autoAug.pl), parameter optimization (autoAugTrain.pl), and genome-wide gene prediction (autoAugPred.pl). Broad protein evidence was elucidated through mini-prot v0.18 alignment of avian proteins (taxonomy ID 8782) from the UniProt database [37]. The initial dataset was filtered using seqkit v2.10.0 to remove fragmented, partial, or exceptionally long sequences (>5,000 amino acids), reducing the dataset to 5,728,600 proteins [38]. Further clustering using CD-HIT v4.6.8 at 90% sequence identity yielded 965,952 bird protein sequences for alignment [39].

All annotation outputs were first converted to EVM-compatible formats using EvmUtils conversion scripts. Final consensus gene models were predicted using EVM with the following weights: AUGUSTUS-raptor=5, GeMoMa=4, mini-prot=2, AUGUSTUS-chicken=1. Annotation quality of all annotation approaches was evaluated using BUSCO in protein mode.

Functional annotation was conducted using eggNOG-mapper v2 to assign Gene Ontology (GO) terms and orthologous groups [40]. The same approach was applied to the three reference Accipitridae species for comparative analysis. GO term distributions across all three GO domains (cellular component, molecular function, and biological process) were assessed with the top 15 GO terms of each species visualized via ggplot2 in R. Shared and unique orthologous groups were visualized using UpSet plots, while protein domains and families were identified using InterProScan v5.75 [41].

Phylogenetic analysis

The *P. jefferyi* draft genome made by our study was compared to the previously submitted reference genome for the *P. jefferyi* in GenBank [19], twenty-one (21) genomes of raptors under the family Accipitridae, as well as two (2) outgroups, the osprey (*Pandion haliaetus*) and the king vulture (*Sarcorampus papa*) belonging to the Pandionidae and Cathartidae families, respectively (Supplementary Table S5). Highly conserved orthologs annotated by BUSCO were extracted for each of the genome assemblies, and those common to all species were used to construct gene alignments using MAFFT v7.5.0 [42, 43] and trimmed for terminal regions using trimAl v1.4.1 [44]. A total of 1,413 trimmed gene alignments were obtained from this and concatenated to generate an overall alignment. The alignment file was then filtered for polymorphic sites such as single-nucleotide polymorphisms (SNPs) and insertion-deletion mutations (indels) and removed ambiguous bases (N's) to construct the final phylip file that was used for phylogeny construction. Lastly, IQ-TREE2 v2.4.0 [45] was used to select for the best nucleotide substitution model and for the construction of the maximum-likelihood phylogeny with

1000 bootstraps. The reconstructed phylogenetic tree was visualized in iTOL [46].

Heterozygosity and demographic history

Individual reads were aligned to the meta-assembly using BWA-MEM2 v2.2.1, with resulting SAM files processed using SAMtools v1.16.1 [47, 48]. Alignment files were further refined using GATK v4.6.1 AddOrReplaceReadGroups and MarkDuplicates modules [49]. Paired-end mapping rates were assessed using samtools flagstat and coverage (Supplementary Tables S6 and S7) [48].

Genome-wide heterozygosity was estimated using genotype likelihood calculations in ANGSD v0.940 [50]. Site allele frequencies were calculated using -dosaf 1 -GL 1 with quality filters (-C 50 -minq 20 -minmapq 30) to minimize spurious variant calls from sequencing errors and mismapped reads. The folded site frequency spectrum was generated using realSFS with -fold 1 -maxiter 100, and heterozygosity for each individual was calculated as the ratio of heterozygous sites (H_1) to total sites [51].

To assess methodological consistency and enable direct comparison with previous raptor studies, heterozygosity was also calculated using hard genotype calls following Zou et al. (2021) [52]. Variants were called using bcftools v1.20 mpileup with -Q 30 and bcftools call with -mv flags, and heterozygous sites (GT = 'het') were counted directly from the resulting VCF files [48]. Both methods yielded highly correlated estimates (Pearson $r > 0.99$; Supplementary Figure S3), with the ANGSD approach producing more conservative values due to probabilistic modeling of genotype uncertainty and stringent quality filtering. Accordingly, the ANGSD-derived estimates were used in this study.

Historical demographic patterns were inferred using Stairway Plot 2 with folded site frequency spectra (SFS) [53]. Effective population size (N_e) changes were estimated assuming a mutation rate of 4.6×10^{-9} per site per generation (derived from collared flycatcher estimates; Smeds et al., 2016, and was used in vultures, Zou et al., 2021, and seagulls, Huynh et al., 2023) and a generation time of 18 years (IUCN Red List data) [7, 52, 54, 55]. Analysis employed the default two-epoch model with 67% sites sampled for training, 200 bootstrap replicates, and singleton exclusion. Historical N_e was estimated using the formula: $\theta = 4N_e\mu$, where mean genome-wide heterozygosity served as a proxy for θ , with the same mutation rate, and mutation-drift equilibrium was assumed [56].

Demographic history was further analyzed using the Pairwise Sequentially Markovian Coalescent (PSMC) method [57]. PSMC and SFS-based methods have complementary temporal resolutions: PSMC resolves ancient

demographic histories (from $\sim 10^4$ to 10^6 years ago) [57, 58], while SFS-based methods like Stairway Plot perform better for recent demographic changes [53, 59]. PSMC analysis was performed on whole-genome sequences from 35 individuals using an atomic time interval of "4 + 30 × 2 + 4 + 6 + 10" with default parameters (-N25 -t15 -r5 -p) and the same mutation rate and generation time ($\mu = 4.6 \times 10^{-9}$, $g = 18$ years) [54, 55].

Results

Genome assembly

The total length of the Philippine eagle genome was estimated to be 1.17 Gbp in Redbean and 1.16 Gbp in Flye. The Redbean and Flye meta-assemblies contained 6,149 and 10,732 contigs, respectively, reducing the average of 36,440 contigs across the 35 *P. jefferyi* short-read assemblies (Supplementary Table S4). The Redbean and Flye had N50 values of 395,632 and 217,643, respectively. Both Redbean and Flye assemblers were also able to eliminate the presence of ambiguous bases (N's) in their meta-assemblies, with zero #N's per 100 kbp for both meta-assemblies. This is in contrast to the short-read assemblies from the Philippine eagle individuals, having an average of 645 ambiguous bases per 100 kbp (Table 1). Overall, quality metrics of meta-assemblies from both methods showed great improvement of the Philippine eagle genome compared to metrics of the short-read assemblies.

BUSCO analysis with the aves_odb10 database showed that the Flye meta-assembly was more complete (95.4%) than Redbean (90.1%). The Flye meta-assembly also had fewer missing BUSCOs (3.2%) than the Redbean one (9.1%). Notably, the Redbean meta-assembly had comparable BUSCO completeness to the average de novo assemblies (90.1% vs 88.0%) and a higher proportion of missing BUSCOs (9.1% vs 5.2%) (Supplementary Table S4). Because of this, the Flye meta-assembly was ultimately chosen as the working reference genome for downstream analysis in this study.

The available reference genome for *P. jefferyi* (GCA_025728025.1) was also compared to our meta-assembly [19]. With fewer contigs and a larger N50, the reference genome is more contiguous than our draft genome meta-assembly. However, upon checking the proportion of unresolved gaps (13,178 #N's per 100 kbp) and BUSCO completeness (85.1%) metrics of the NCBI reference genome, we see a great improvement in completeness in our draft genome, which is free of unresolved base calls and has a 95.4% BUSCO completeness. This also makes it more amenable for use in gene annotation and downstream analyses [60].

RepeatMasker masked 59,132,847 bases (5.08%) of the genome, with interspersed repeats comprising the majority (Fig. 1). Long interspersed nuclear elements (2.41%),

Table 1 Philippine eagle (*P. jefferyi*) genome assembly metrics. Best short-read assembly metrics refer to the individual assembly with the highest N50 (PEF-12). Average short-read assembly metrics refer to the average values for each of the 35 *P. jefferyi* de novo assemblies generated in this study. The meta-assembly column shows the genome assembly metrics for the Flye-assembled genome (this paper). Both metrics were compared with the *P. jefferyi* reference genome available in NCBI [19]

Assembly Metrics	Best Short-read assembly	Average Short-read assembly	Meta-assembly	Reference genome
# of contigs (>= 500 bp)	25,591	36,440	10,732	1,135
Largest contig	602,466	461,899	1,384,840	84,365,990
Total genome length	1,156,765,217	1,158,184,517	1,162,923,170	1,247,235,770
GC (%)	41.71%	41.63%	41.56%	41.37%
N50	82,107	56,797	217,643	38,333,288
# N's per 100 kbp	735.81	645.29	0	13,178.38
Complete BUSCOs	91.5%	88.0%	95.4%	85.1%
Single-copy BUSCOs	91.2%	87.6%	94.9%	84.8%
Duplicated BUSCOs	0.3%	0.4%	0.5%	0.3%
Fragmented BUSCOs	4.4%	6.8%	1.4%	6.7%
Missing BUSCOs	4.1%	5.2%	3.2%	8.2%

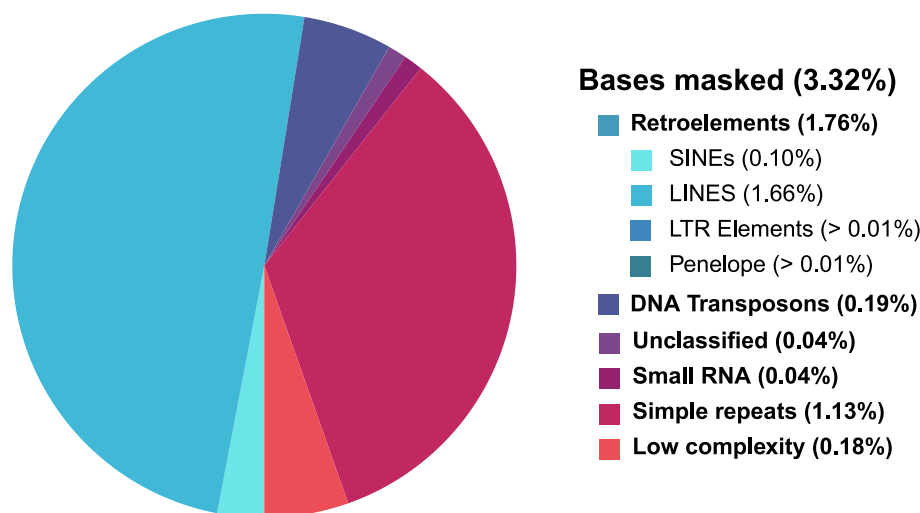


Fig. 1 Repeat elements identified by repeat masking in the draft genome of *P. jefferyi*. Given are the proportions of bases masked as repeats in six categories (Retroelements, DNA Transposons, Unclassified repeats, Small RNA repeats, Simple repeats, and Low complexity repeats)

long terminal repeats (1.28%), and simple repeats (0.89%) accounted for the largest fractions of masked bases, alongside smaller contributions from DNA transposons (0.20%) and low complexity repeats (0.17%).

Genome Annotation

Transfer RNA annotation using tRNAscan-SE identified a total of 191 tRNA genes confirmed by Infernal, comprising 170 canonical tRNAs, 17 predicted pseudogenes, 3 selenocysteine tRNAs, and 1 unclassified tRNA. Notably, 20 of the 64 possible anticodons were absent, including both ACA and GCA codons for cysteine, indicating a lack of corresponding tRNA genes for this amino acid. Ribosomal RNA annotation via barrnap revealed five rRNA features: partial sequences of 18S (27% alignment), 28S (31% alignment), and 5S (73% alignment), along with two complete 5S rRNA genes. No 5.8S rRNA genes were detected.

The final gene set produced by EVM comprises 24,781 genes (Table 2), spanning a total length of 347,818,533 base pairs and covering 29.90% of the draft assembly, with a gene density of approximately 21.30 genes per megabase. BUSCO analysis of the protein sequences revealed 93.0% completeness, with 2.8% fragmented and 4.2% missing genes, an improvement over the best pre-EVM annotation, GeMoMa, which showed 90.2% completeness, 1.4% fragmented, and 8.4% missing. In addition to increasing the number of complete BUSCOs, the final gene set significantly reduced the proportion of duplicate complete orthologous genes from 36.4% to just 0.5%, indicating a more refined and less redundant annotation. The higher duplication rate in GeMoMa is likely due to its use of homology-based predictions from three different Accipitridae raptor species.

Out of the 24,781 genes identified, eggNOG-map-per functionally annotated 15,072 genes (60.8%), while

Table 2 Structural annotation of protein-coding genes of *Pitheophaga jefferyi*. Statistics including BUSCO completeness metrics (protein mode) for the final EvidenceModeler (EVM) consensus annotation compared with contributing annotations: GeMoMa (homology-based), Aug-Raptor (AUGUSTUS; model trained on raptors), and Aug-Chicken (AUGUSTUS; model trained on chicken). The EVM annotation integrates evidence from all three methods to produce the final gene set used in subsequent analyses

	EVM	GeMoMa	Aug-Raptor	Aug-Chicken
Total Gene Count	24,781	30,404	27,822	46,461
Total Gene Length (bp)	347,818,533	772,824,705	366,529,379	426,324,956
Mean Gene Length (bp)	14,035	25,418	13,174	9,175
Gene Density (genes/Mb)	21.30	26.14	23.92	39.95
Gene Coverage	29.90%	66.45%	31.51%	36.65%
Complete BUSCOs	93.0%	90.2%	83.9%	76.1%
Single-copy BUSCOs	92.5%	53.9%	83.3%	75.5%
Duplicated BUSCOs	0.5%	36.4%	0.6%	0.6%
Frag-mented BUSCOs	2.8%	1.4%	6.5%	10.1%
Missing BUSCOs	4.2%	8.4%	9.6%	13.8%

InterProScan detected functional domains in 20,887 genes (84.3%). GO annotations at level 5 (Fig. 2A), highlighted the most frequently assigned terms across the three major GO categories: for Cellular Component, “intracellular membrane-bounded organelle” (65.4%), “nuclear lumen” (14.6%), and “cytoskeleton” (10.2%); for Molecular Function, “DNA binding” (14.1%), “kinase activity” (7.9%), and “regulatory region nucleic acid binding” (7.6%); and for Biological Processes, “cellular protein metabolic process” (21.5%), “regulation of gene expression” (20.1%), and “regulation of nucleobase-containing compound metabolic process” (18.9%); all reflecting evolutionarily conserved functions.

These top terms were consistent with those found in other raptor proteomes. However, the Philippine eagle

showed notably lower representation of “cation transmembrane transporter activity,” “inorganic cation transmembrane transporter activity,” and related ion channel functions. EggNOG-mapper also assigned 13,467 orthologous groups (OGs) to *P. jefferyi*, with approximately 14,300 OGs in the reference proteomes; of these, 12,809 OGs (about 95%) were shared, while 1,057 OGs present in the reference proteomes were absent in *P. jefferyi*, indicating some missing core raptor annotations (Fig. 2B).

Interestingly, *P. jefferyi* had 152 unique OGs, more than the 30, 61, and 28 unique OGs found in the other raptors. InterProScan further identified protein families and domains from various databases, shown in Fig. 2C, with the most abundant being “Zinc finger C2H2-type” (IPR013087), “EGF-like domain” (IPR000742), and “WD40 repeat” (IPR001680), all of which are evolutionarily conserved.

Phylogenetic analysis

The phylogenetic tree composed of the *P. jefferyi* draft genome, the currently available *P. jefferyi* reference genome (GCA_025728025.1), and twenty-three (23) other raptors showed the clustering of four subfamilies in the Accipitridae family (Buteoninae, Accipitrinae, Aquilinae, Circaetinae), as well as the correct classification of the osprey (*Pandion haliaetus*) and king vulture (*Sarcophaga papia*) as outgroups (Fig. 3). Phylogenetic placement of this study's *P. jefferyi* draft genome is within the snake eagle subfamily Circaetinae, along with the short-toed snake eagle (*Circaetus gallicus*), western banded snake eagle (*Circaetus cinerascens*), black-chested snake eagle (*Circaetus pectoralis*), and Congo serpent eagle (*Circaetus spectabilis*). This is consistent with the taxonomic revisions made by the International Ornithological Congress (IOC) [61] as well as existing literature on Accipitridae taxonomy [62–68]. The cinereous vulture (*Aegypius monachus*) genome is also placed adjacent to this clade. Another major clade was formed by the Aquilinae family, composed of the mountain hawk-eagle (*Nisaetus nipalensis*), ornate hawk-eagle (*Spizaetus ornatus*), black hawk-eagle (*Spizaetus tyrannus*), and Wahlberg's eagle (*Hieraetus wahlbergi*), and the true eagles (genus *Aquila*). The Eurasian goshawk (*Astur gentilis*), Eurasian sparrowhawk (*Accipiter nisus*), brown goshawk (*Accipiter fasciatus*), and besra sparrowhawk (*Accipiter virgatus*) clustered together and are all members of the subfamily Accipitrinae, also known as true hawks [66–68]. Finally, the white-tailed eagle (*Haliaetus albicilla*) and the bald eagle formed a cluster with the rufous-winged buzzard (*Butastur liventer*) consistent with the Buteoninae subfamily taxonomy. Interestingly, the Blyth's hawk-eagle (*Nisaetus alboniger*) and the Philippine Eagle reference genome (GCA_025728025.1) were also placed under this major clade, despite empirical evidence that

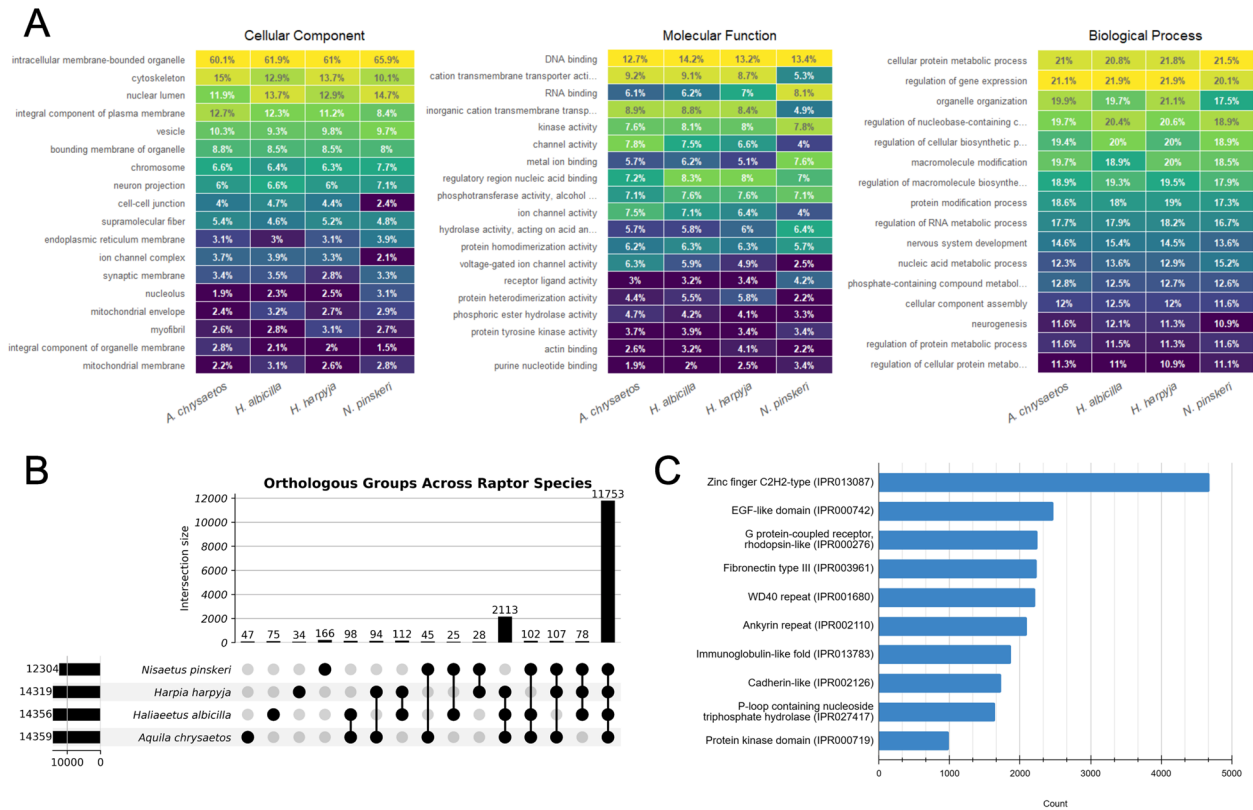


Fig. 2 Summary of functional annotation of the Philippine eagle compared to reference species. **A** Top GO term distributions at Level 5 across *A. chrysaetos*, *H. albicilla*, *H. harpyja*, and *P. jefferyi*, grouped under Cellular Component, Molecular Function, and Biological Process. **B** UpSet plot illustrating shared orthologous groups (OGs) among the four raptors. The bar graph counts group overlap, while the dot matrix indicates all possible species combinations. **C** Horizontal bar chart of the most represented InterProScan protein families

these raptors belong in subfamilies *Aquilinae* and *Circaeinae*, respectively [62, 64].

Genetic Diversity

The average genome-wide heterozygosity across all sampled Philippine eagles is estimated at 0.000309. When compared across species with different conservation statuses, heterozygosity values were consistently lower in species of higher threat (IUCN) categories, with Near Threatened, Vulnerable, Endangered, and Critically Endangered species showing significantly reduced levels relative to Least Concern taxa (Fig. 4A). Among these, the Philippine eagle exhibited one of the lowest heterozygosity values compared to the other Critically Endangered species.

Focusing on raptors (from families belonging to avian orders Accipitriformes, Strigiformes, Falconiformes, Cathartiformes, and Cariamiformes), heterozygosity values were generally low across all conservation statuses, including those species listed as Least Concern (Fig. 4B) [69]. The Philippine eagle has one of the lowest genome-wide heterozygosity compared to raptors with available heterozygosity estimates. Among the individual eagles

sampled, genome-wide heterozygosity reached as low as 0.000207 (Fig. 4C).

To validate our genome-wide heterozygosity estimates and enable comparison with other published values, we calculated genome-wide heterozygosity using three approaches: our primary ANGSD method with stringent quality filters (mean=0.000309), ANGSD without filters (mean=0.000487), and the hard-call bcftools approach from Zou et al. (2021) (mean=0.000461) [52]. All three methods showed extremely high correlation (Pearson $r > 0.99$ for all pairwise comparisons; Supplementary Figure S3), with almost identical ranking of individuals by heterozygosity. Even with its highest average heterozygosity estimates, the Philippine eagle remained one of the lowest among the raptors, only slightly above *H. albicilla* (0.000400) and *H. leucocephalus* (0.000430) [69]. The filtered ANGSD approach yielded the most conservative estimates, which are approximately 33% lower than the unfiltered approach and 49% lower than the hard-call method. Apart from the removal of low-quality variants in the filtered ANGSD approach, we chose to report the estimates of the said approach as these reflect the “worst-case” scenario for species conservation.

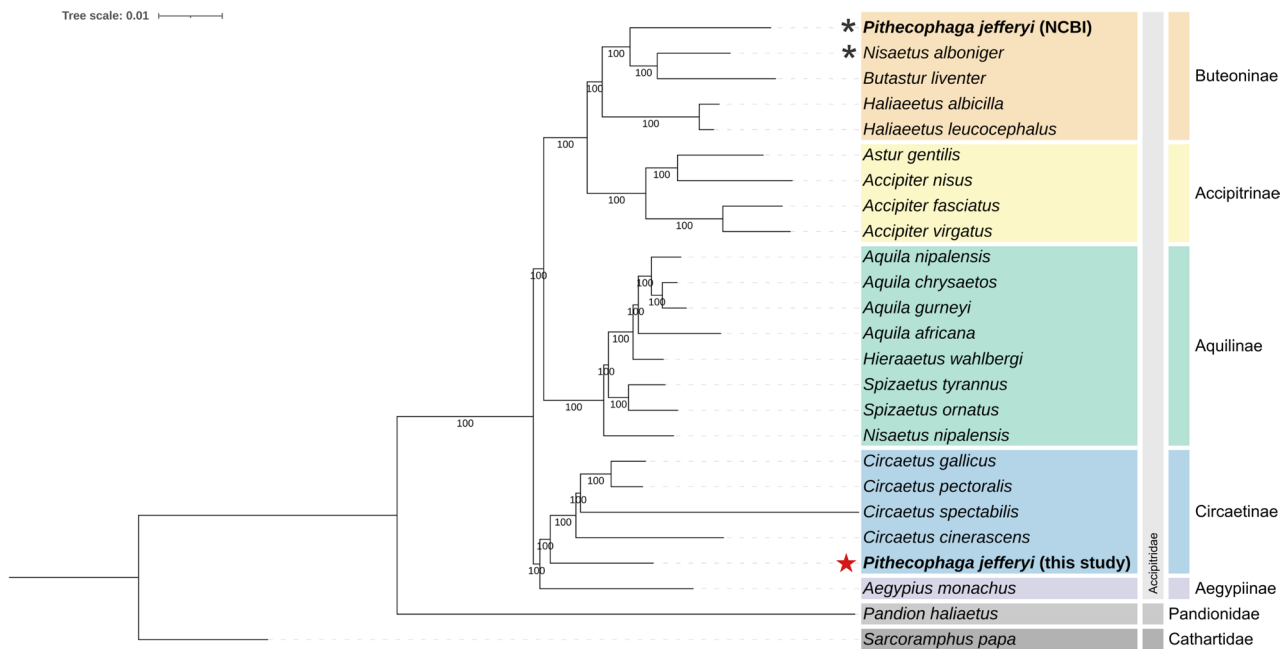


Fig. 3 Phylogenetic tree of Accipitriformes including the newly assembled *Pithecopaga jefferyi* genome. Families (Cathartidae, Pandionidae, and Accipitridae) are indicated, with subfamilies within Accipitridae (Aegypiinae, Circaetinae, Aquilinae, Accipitrinae, and Buteoninae) distinguished by color. The *P. jefferyi* genome assembled in this study is marked with a red star. Misplacements of the *P. jefferyi* reference genome (NCBI, GCA_025728025.1) and *Nisaetus alboniger* (GCA_025447895.1) within Buteoninae are denoted by asterisks. Bootstrap support values are shown at the nodes. The phylogeny was inferred from a phylip file derived from polymorphic sites (excluding N's) of concatenated alignments of 1,413 universal single-copy orthologs, with *Sarcoramphus papa* (Cathartidae) designated as the outgroup

Demographic history

The inferred historic effective population size from the average genome-wide heterozygosity was 16,795, drastically larger than any population estimates today [4]. The Stairway plot, assuming the same mutation rate and a generation time of 18 years, suggested two major historical bottlenecks followed by an ongoing population decline (Fig. 5). The earliest bottleneck occurred approximately 800,000 years ago, reducing the effective population size from ~40,000 to 10,000 individuals. The population subsequently recovered to ~100,000 individuals 400,000 years later. A second, more severe bottleneck began around 80,000 years ago, reducing N_e to its second lowest point of ~4,000 individuals at approximately 50,000 years ago. Following this bottleneck, the population recovered to ~70,000 individuals by 30,000 years ago. More recently, an ongoing population decline that began approximately 3,000 years ago led to N_e decreasing from ~50,000 to an estimated ~2,000 individuals 100 years ago.

The PSMC analysis (Supplementary Figure S4) reconstructed demographic history from 10^4 to 10^7 years ago, whereas the Stairway plot captured demographic history from 10^2 to 10^6 years ago. Both methods detected the first decline and recovery at around 10^6 years ago. However, while the PSMC analysis captured the second decline, it did not resolve the subsequent recovery at

around 40,000 years ago. PSMC trajectories were largely consistent among Mindanao individuals ($n = 35$) from 10^7 to ~300,000 years ago. After this point, the single Samar individual showed a more rapid and severe population decline and maintained substantially lower N_e estimates compared to the Mindanao population thereafter. However, this observation is based on a single Samar sample and requires confirmation with additional individuals.

Discussion

Quality and completeness of the Philippine eagle draft assembly

The genome assembly for *P. jefferyi* generated in this study demonstrates improved quality compared to the previous effort, particularly when considering assembly methods, the impact of fragmented DNA, and the benefits of combining multiple assemblies into a consensus meta-assembly.

The meta-assemblies showed significant enhancements in contiguity and completeness and eliminated the presence of ambiguous bases (Ns) compared to individual assemblies. This approach increased genome coverage and effectively resolved ambiguous base calls, and signifies that unresolved regions in one individual assembly were complemented by contigs from other assemblies where those regions were resolved [70–72]. The final meta-assemblies also improved contiguity, indicated by

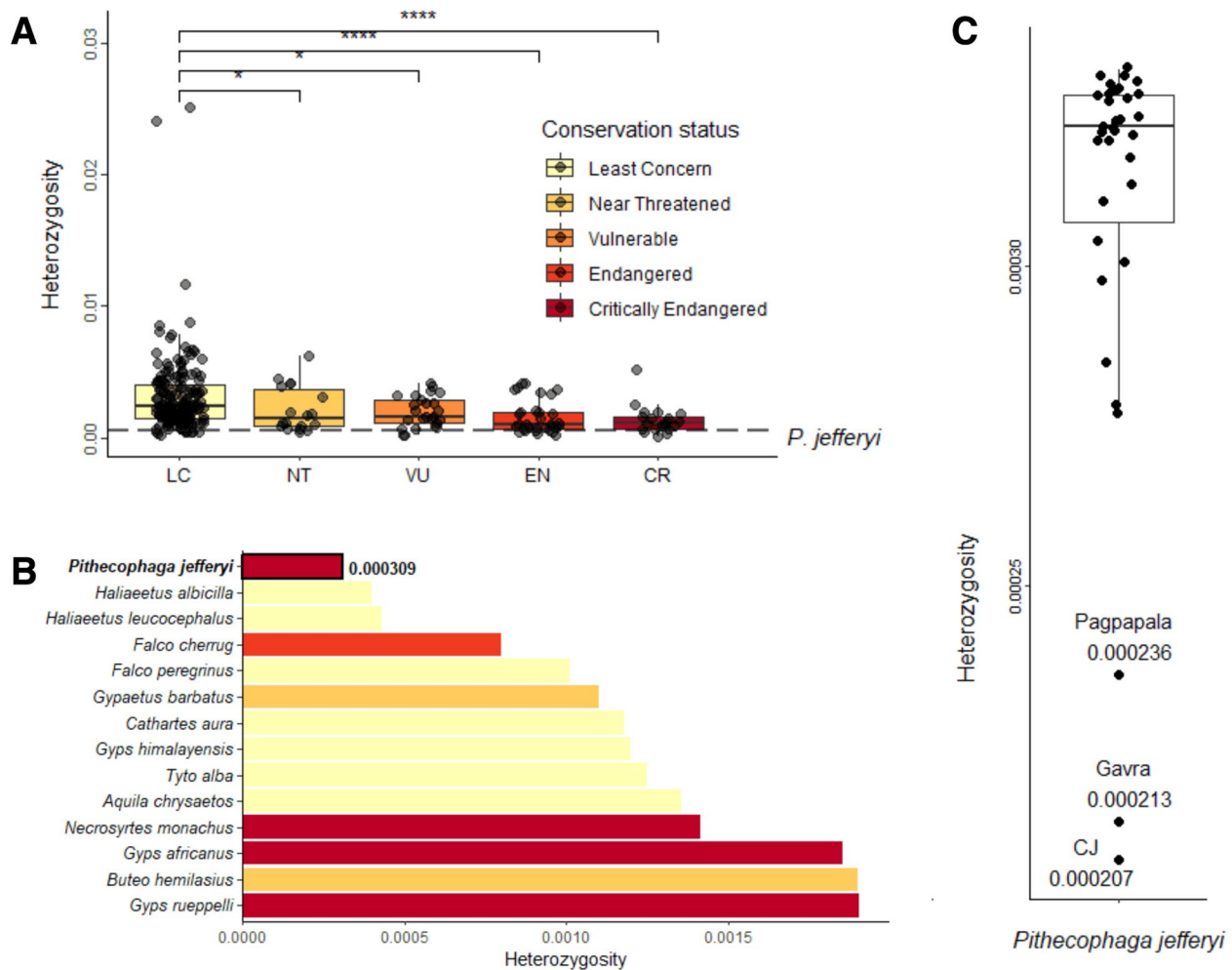


Fig. 4 Genome-wide heterozygosity of the Philippine eagle relative to other threatened species. **A** Boxplot distribution of genome-wide heterozygosity coefficients for threatened species, grouped by conservation status [51, 69]. The heterozygosity of the Philippine eagle is demarcated with a dashed line. **B** Genome-wide heterozygosity of the Philippine eagle relative to other threatened raptors (Accipitriformes, Strigiformes, Falconiformes, and Cariamiformes) [52]. **C** Individually calculated genome-wide heterozygosity for each eagle sample

higher N50 values, compared to the initial short-read assemblies.

The draft genome of *P. jefferyi* produced in this study demonstrates higher completeness and lower ambiguity when directly compared to the reference genome previously published in NCBI, which was found to have a high proportion of unresolved gaps (N's per 100 kbp), despite having a higher overall N50 value than the meta-assembly. The assembly was reportedly based on DNA extracted from a toe pad sample, which is generally recognized as being more degraded than DNA obtained from fresh tissue [73, 74]. This might have caused the high proportion of ambiguous base calls in the reference assembly, resulting in a lower BUSCO completeness score.

Conversely, the present study utilized DNA extracted from blood samples of live Philippine eagles, providing higher-quality input material with detailed

documentation. Consequently, apart from the elimination of ambiguous base calls, the *P. jefferyi* draft genome from this study boasts a BUSCO completeness of 95.4%. These strong metrics represent a significant improvement for critical downstream applications like gene annotation and diversity analysis.

While the draft genome lays essential groundwork, its limitations highlight clear opportunities for future refinement through long reads and Hi-C sequencing to achieve chromosome-level resolution and transcriptomic validation to refine gene models and functional annotations. The difficulty in resolving repetitive regions in short read assemblies has led to incomplete annotation of rRNA genes, particularly 18S, 28S, and the missing 5.8S, due to their location in highly repetitive nucleolar organizer regions, while some of the shorter and more dispersed 5S rRNA were recovered [75]. The genome shows only 5.08% masked repeat sequences, significantly lower than

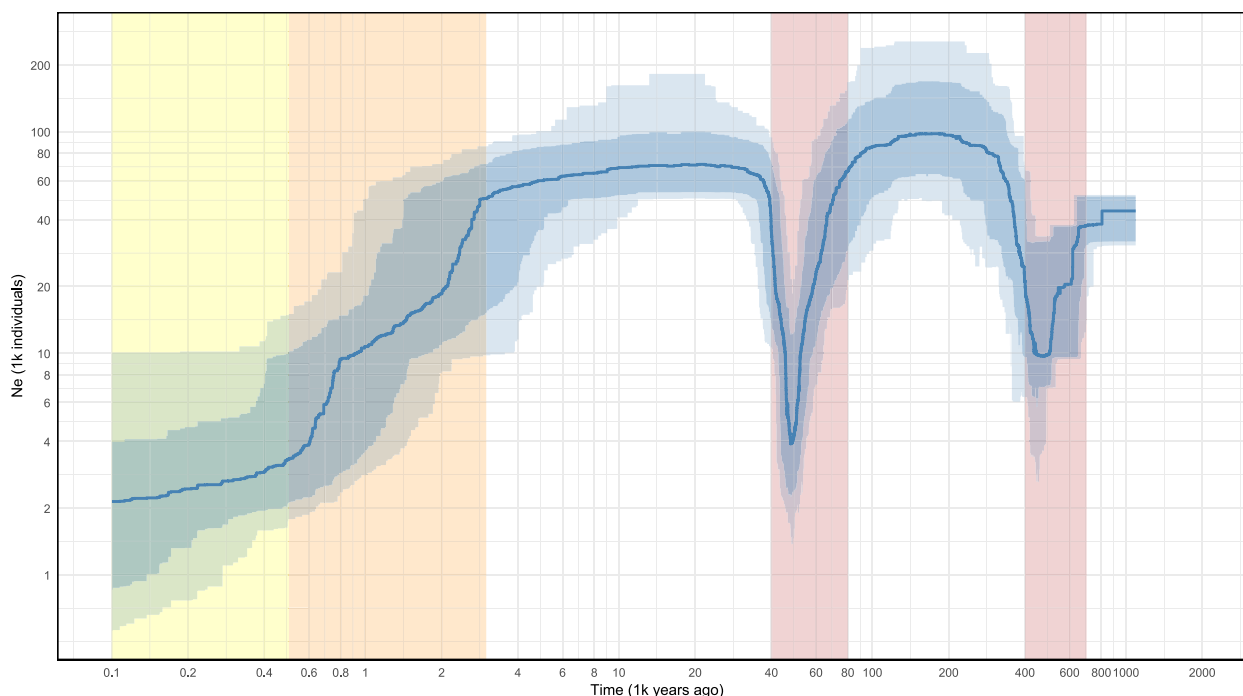


Fig. 5 Stairway plot of the demographic history of the Philippine eagle. The dark blue line represents the median estimate of effective population size over time. Blue and light blue shaded areas indicate the 87.5%–12.5% and 97.5%–2.5% confidence intervals, respectively. The red boxes highlight two population bottlenecks, the orange box highlights the start of the ongoing decline, and the yellow box highlights the start of deforestation in the Philippines

the 16.33% in *H. harpyja*, likely due to collapsed repetitive regions inherent to short-read sequencing [76].

Functional annotation reveals a deficit of 1,057 orthologous groups (OGs) compared to reference raptors, suggesting an underestimation of the true gene set, though conserved biological pathways appear adequately represented. The 152 unique OGs may reflect either lineage-specific traits or false predictions from assembly fragmentation, which warrants further investigation [77]. While homology-based and ab initio methods provide a comparative framework, transcriptomic data will be essential to validate gene function, identify pseudogenes, and uncover splicing diversity [78–80]. Despite these limitations, this draft genome forms a foundation for conservation genomics in Philippine raptors, and can enable studies on population structure, adaptive evolution, and genetic monitoring, critical for managing this apex predator and, indirectly, the country's rich biodiversity [81–83].

Phylogenetic placement of the *P. jefferyi* draft genome in the Accipitridae family

This study found that the *P. jefferyi* draft genome formed a clade with raptors in the subfamily Circaetinae and the cinereus vulture (*A. monachus*) [84]. This is consistent with IOC standards [61] and several prior molecular phylogenetic studies [62–64].

The Circaetinae subfamily includes serpent eagles (*Spirolornis*, 6 species), bateleur (*Terathopius ecaudatus*), snake eagles (Circaetus, 7 species), and the Madagascan serpent eagle (*Eutriorchis ater*) [61]. This relationship of the draft genome (this study) with Circaetinae supports the taxonomic origins of the extremely large *Pithecophaga* to be derived from smaller ancestral snake eagles. Hence, supporting the result of studies on the diet of *P. jefferyi*, which comprises 25% snakes and up to 37% reptiles [65]. This taxonomic arrangement also confirms that *P. jefferyi* is not related to harpy eagles belonging to the subfamily Harpiinae, which includes the enormous harpy eagle and Papuan harpy eagle (*Harpyopsis novaeguineae*).

Apart from the empirically consistent placement of the Aegypinae and Circaetinae raptors as sister groups, the phylogeny also correctly clustered the booted eagles (Aquilinae), which include raptors belonging to the *Nisaetus*, *Spizaetus*, *Hieraetus*, and *Aquila* genera [62, 64, 66]; Buteonine or broad-winged hawks, kites and allies (subfamily Buteoninae), which consist of the *Butastur* and *Haliaetus* raptors [64, 67]; and goshawks (Accipitrinae), consisting of the *Accipiter* and *Astur* raptors [66–68]. It also correctly placed the osprey and king vulture as outgroups to the rest of the Accipitridae species. The presence of these empirically consistent phylogenetic placements reinforces the robustness of using highly-conserved avian gene orthologs in the analysis.

There were also phylogenetic inferences based on the reference genomes for *P. jefferyi* (GCA_025728025.1) and *N. alboniger* (GCA_025447895.1) that resulted in an alternative, known conflicting topology, placing them within the Buteoninae subfamily. This observed topological discordance may reflect challenges in genome assembly and scaffolding when utilizing short-read sequences derived from highly fragmented DNA, particularly in repeat-rich regions [66, 85, 86], and imply that the quality of these available genome assemblies warrants further assessment. The robust placement of *P. jefferyi* into the Circaetinae clade achieved by our draft genome successfully resolves this phylogenetic inconsistency, consistent with established molecular evidence [61–64]. Despite a simple BUSCO-based comparative phylogenetic methodology, the empirically consistent placement of the *P. jefferyi* draft genome indicates that the assembly may be a more accurate representation of the biological reality of the Philippine eagle genome compared to the available NCBI reference.

Demographic history reveals ancient bottlenecks and a recent ongoing decline

The effective population size (N_e) provides insight into the demographic history and genetic health of a species, representing the number of individuals in an idealized population that would experience the same rate of genetic drift as the observed population [87]. In this study, the N_e was inferred using the site frequency spectrum, which estimates a long-term historical average over many generations. Accordingly, the resulting N_e represents the size of a hypothetical stable population maintained throughout the species' existence, rather than its current demographic status.

For the Philippine eagle, the inferred historic effective population size in this study was approximately 17,000, which is larger than current census population estimates of around 392 breeding pairs [4]. This large difference between the effective and census (N_c) population size suggests that the population is undergoing a relatively recent and steep decline. While N_e is generally expected to be lower than N_c in stable populations, owing to factors such as unequal reproductive success and non-random mating, the reversal observed here suggests a demographic contraction from a historically larger, more genetically diverse population [88]. In effect, the elevated N_e serves as genetic evidence for a trend already recognized through field observations – the Philippine eagle population is shrinking and remains at significant risk.

Similar patterns have been observed in other raptor species, such as golden eagle subpopulations in Japan and Scotland, where historical N_e estimates of ~70,000 likely predate population isolation and reflect the effective size of larger, geographically proximate continental

populations before fragmentation [89]. For the Philippine eagle, which is found across multiple islands (Luzon, Leyte, Samar, and Mindanao), the high historical N_e estimate likely corresponds to a time before island fragmentation and subsequent population decline. It should be noted that the samples analyzed in this study were predominantly from Mindanao, with one individual from Samar; therefore, they may not fully capture the genetic diversity across all subpopulations, particularly those from Luzon and Samar. The N_e estimate likely reflects a long-term average that predates the current fragmented island distribution and recent population contractions.

The temporal coincidence between the most recent bottleneck (~80,000 years ago, reducing N_e to ~4,000 individuals) and the Youngest Toba Eruption (~74,000 years ago) warrants consideration. The eruption happened at what is now Lake Toba in Sumatra, Indonesia, which lies in roughly the same geographical region as the present distribution of the Philippine eagle [90]. While the spatial and temporal coincidence of this volcanic eruption with the inferred bottleneck suggests a possible association, it does not provide definitive evidence of causation. The Youngest Toba Eruption was initially hypothesized to have caused global cooling, volcanic winter, and population bottlenecks across various species, including humans [91]. However, more recent studies suggest that the eruption's effects on global climate may have been overestimated [90]. This raises the possibility that factors beyond the Toba eruption may have contributed to the Philippine eagle's demographic decline during this period.

The most recent bottleneck fits within broader avian demographic trends during the Last Glacial Period (LGP; 110,000 years ago). Demographic reconstructions using whole genomes and PSMC analysis revealed that many bird species experienced significant effective population size reductions during this period, with 22 of 38 analyzed species showing notable declines [92]. Raptors like the white-tailed eagle, bald eagle (*Haliaeetus leucocephalus*), barn owl (*Tyto alba*), and turkey vulture (*Cathartes aura*) exhibited sustained population collapses, with the eagles declining from pre-LGP levels of ~60,000 individuals to post-glacial lows of only 2,000–6,000 individuals by 10,000 years ago [92, 93].

In contrast, however, the Philippine eagle's effective population size (N_e) rebounded to approximately 70,000 individuals as early as 30,000 years ago, approaching pre-LGP estimates and exhibiting a recovery trajectory not observed in other raptors [92]. This demographic resurgence may be partially attributed to the formation of the Greater Mindanao faunal region or the Mindanao Pleistocene Aggregate Island Complex (PAIC) during the LGP, when sea levels fell by approximately 120–130 m. This eustatic drop exposed extensive continental shelves and

land bridges that temporarily unified Mindanao, Samar, Leyte, Bohol, Dinagat, and adjacent satellite islands into a single contiguous landmass, as delineated by Heaney's faunal regions and the PAIC model using the 120-m bathymetric line [94–96]. These resulting land aggregates substantially expanded the total land area of the Philippine archipelago — potentially reaching ~350,000–400,000 km² — thereby increasing habitat availability, potentially enhancing prey abundance, and facilitating gene flow among subpopulations that had been isolated during previous interglacial periods. Notably, even during the Last Glacial Maximum (~22,000 years ago), N_e estimates indicate that the Philippine eagle population persisted, in contrast to the sustained declines observed in other avian populations, suggesting a degree of resilience to climatic challenges, potentially mediated by the ecological buffering effects of PAIC-driven habitat aggregation and landmass expansion.

The PSMC analysis temporally complemented the Stairway plot, reconstructing demographic history from 10⁴ to 10⁷ years ago compared to the Stairway plot's 10² to 10⁶ year range. Both methods were fairly consistent in more ancient time ranges, detecting the first major decline and recovery at ~10⁶ years ago and the subsequent second major decline at around 100,000 years ago. However, PSMC failed to resolve the recovery following the second bottleneck, which is consistent with the method's reduced power for detecting demographic changes in the recent past [59]. The single Samar individual exhibited a divergent demographic trajectory beginning ~300,000 years ago, with substantially lower N_e estimates compared to Mindanao individuals, though this observation requires confirmation with additional samples. While broadly consistent with the Stairway plot in deeper time ranges, PSMC was unable to resolve demographic shifts in the most recent millennia, underscoring the complementary strengths of the two approaches [58, 59].

Although the species demonstrated resilience during earlier climatic fluctuations, its demographic history over the past 3,000 years reflects a prolonged and ongoing population decline, suggesting that the hypothesized ecological advantages conferred by the Greater Mindanao faunal region or Greater Mindanao PAIC had since been lost, as Holocene sea levels disaggregated these landmasses by ~7,000 years ago [97]. This reduction in landmass and loss of habitat connectivity, compounded by emerging stressors such as early anthropogenic pressures, may have disrupted population stability. Effective population size estimates indicate a drop from ~50,000 to ~2,000 individuals between 3 kya and 0.1 kya, which is faster and more severe than earlier bottlenecks. While deforestation is a recognized anthropogenic threat, it can only partially explain the timing of the population

decline. Spanish colonization of the Philippines began 500 years ago, when forest cover was still at 90%, declining to 71% by 100 years ago (around 1900), coinciding with the endpoint of our N_e estimates [14, 98]. Forest cover has since been drastically reduced to only 23% (72,000 km²) by 2002, of which less than 10,000 km² are considered primary forests [99, 100]. This timeline reveals that a significant population decline was already well underway even before major deforestation began, suggesting the influence of additional stressors, such as ecological, climatic, or other anthropogenic causes. Notably, the onset of this decline also coincides with the Austronesian expansion into the Philippine archipelago (~2200–1000 BCE), a period characterized by increasing human population densities, intensified hunting activities, and heightened competition for key eagle prey species such as wild pigs, deer, and macaques. These ecological pressures may have contributed significantly to the species' long-term population contraction, suggesting another potential anthropogenic influence on the population history of the Philippine eagle. [101, 102].

This dissonance between known anthropogenic threats and the timing of population decline stands in stark contrast to the demographic trajectory of other critically endangered birds, such as the crested ibis (*Nipponia nippon*), whose recent population collapse due to overhunting and habitat loss was clearly captured in demographic reconstructions [69]. Unlike the Philippine eagle, whose decline spans millennia, the crested ibis experienced a rapid reduction from ~10,000 individuals to near extinction within the last 100 years, closely aligned with documented anthropogenic pressures. Although our Stairway Plot cannot resolve N_e trends beyond ~100 years ago and therefore cannot determine whether populations have stabilized since conservation efforts began in 1969, the Philippine eagle's long-term decline remains clearly detectable [11]. Recent threats such as habitat loss and human persecution are well documented, yet the prolonged nature of the Philippine eagle's decline raises the possibility that older, less understood ecological or anthropogenic pressures may have also played a role.

Extremely low genetic diversity and drift debt threaten long-term viability

The genomic diversity estimates presented in this study are based on samples predominantly originating from Mindanao, which almost reflects the current distribution of the species rather than a sampling deficiency. Mindanao is recognized as the primary stronghold of the Philippine eagle, hosting the majority of the remaining wild population, with smaller populations in other parts of its historically documented range (Samar, Leyte, and Luzon) [4]. With this context in mind, the following discussion interprets the genomic diversity estimates as reflective

of the genetic diversity of the primary population of the Philippine eagle.

The genomic consequences of the ongoing long-term demographic decline are evident in the Philippine eagle's exceptionally low genome-wide heterozygosity (GWH). Among all raptors with reported GWH estimates, the Philippine eagle exhibits the lowest value, at just 0.000309. Although low heterozygosity alone does not determine extinction risk, since some species may persist with low diversity and others may be endangered despite moderate heterozygosity (see Fig. 4A), this value is concerning when interpreted alongside the species' sustained small population size [103]. With an estimated 392 breeding pairs remaining in the wild, and some more pessimistic assessments suggesting as few as 64 individuals, prolonged genetic drift and limited opportunities for gene flow have likely contributed to the erosion of genetic variation [4, 104]. In this context, its low heterozygosity aligns with well-documented demographic decline and ecological constraints. As such, it serves as a genomic signal of heightened risk, with important implications for the species' persistence and adaptive potential.

In small populations, reduced genome-wide heterozygosity indicates a history of strong genetic drift, which reduces the efficacy of purifying selection against mildly deleterious alleles [105]. Theoretical expectations also suggest that limited mate choice in such populations increases the likelihood of inbreeding, which can elevate homozygosity and expose recessive harmful variants [106]. Although inbreeding was not directly assessed in this study, these dynamics are consistent with the observed low heterozygosity and raise concerns about potential inbreeding depression, reduced fitness, and long-term viability [105, 107]. Genomic resources presented in this study may inform future breeding and management programs, with the protection of the remaining genetic variation being critical to their success. Conservation efforts must extend beyond short-term demographic recovery to ensure both genetic resilience and long-term persistence.

While the Philippine eagle requires short-term intensive population recovery efforts, its comparatively low heterozygosity raises questions about whether demographic gains alone will ensure long-term persistence. As the largest island-endemic raptor, a combination of traits predicting elevated conservation status, the species faces compounded challenges in achieving genetic resilience [108]. Comparative examples from both large continental raptors and island-endemic species illustrate this pattern of genetic vulnerability despite demographic recovery. Large continental eagles such as the white-tailed eagle and the bald eagle were downlisted to Least Concern following population recovery from

DDT-driven declines, yet both retain low genome-wide heterozygosity [109–112]. Among island-endemic raptors, the endangered Mauritius kestrel (*Falco punctatus*) recovered from a near-extinction event of four remaining individuals to now around 140 individuals; despite demographic recovery, genomic analysis from a recent preprint revealed it to have critically low heterozygosity and evidence of sustained recent inbreeding [113–115]. More broadly, island-endemic birds show similar patterns: the Chatham Island robin (*Petroica traversi*) was downlisted from Endangered to Vulnerable following recovery from a single breeding pair, and consequent genomic analysis revealed substantial decreases in heterozygosity and increases in F_{ROH} from historical samples [116–118]. A similar trajectory is observed in the Seychelles magpie-robin (*Copsychus sechellarum*), an island-endemic passerine that recovered from a severe bottleneck of just 12 individuals in the 1960s through translocation-based conservation efforts and was downlisted from Critically Endangered to Endangered in 2005, despite the absence of comprehensive genetic data at the time [119]. Subsequent genomic analysis in 2021 measured its genome-wide heterozygosity at 0.000150, around half of that of the Philippine eagle, and observed long runs of homozygosity, indicating recent inbreeding [120]. More troublingly, they noted that funding and research efforts plateaued following the downlisting, demonstrating how premature declarations of conservation success can compromise long-term protection when genetic factors are overlooked.

These examples demonstrate that demographic recovery does not erase the genomic consequences of bottlenecks. The Philippine eagle faces the additional challenge of 'drift debt' – even with population size recovery, the species would remain genetically imperiled due to genomic erosion from past bottlenecks, with genetic drift effects persisting for many generations [93, 121]. For the Philippine eagle, genetic considerations must be integrated in any future conservation assessment to prevent similar outcomes to the Seychelles magpie-robin, where downlisting preceded comprehensive genetic analysis and led to reduced conservation investment, leaving the species genetically exposed to future threats.

Conclusion

We have successfully constructed and annotated a genome assembly for the Philippine eagle that improves on the completeness and genome coverage of the previously published reference genome. Phylogenetic analysis then confirmed its placement within the subfamily Circaetinae, something the previous reference genome was not able to reflect. This clade placement with snake eagles

agrees with the taxonomic revisions of the International Ornithological Congress.

By using this genomic resource, we found that the Philippine eagle has one of the lowest genome-wide heterozygosities, at an average of 0.000309, compared to raptors and other threatened species. We also found that the Philippine eagle has gone through two population bottlenecks and an ongoing population decline that predates deforestation in the Philippines. Its low genome-wide heterozygosity limits its adaptive capacity and, combined with its dwindling population, makes the Philippine eagle one of the most vulnerable species to ongoing anthropogenic and environmental pressures. Conservation efforts could utilize this genomic resource in monitoring the success of captive breeding and translocation programs.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12864-026-12859-9>.

Supplementary Material 1: Table S1. Metadata of sampled Philippine eagles, including sex, hatch or rescue date, province and region of origin, and known family (directly related individuals). Table S2. General sequencing reads quality statistics of the 35 Philippine eagle samples. Table S3. Relevant QAST metrics of the Flye meta-assembly compared to the individual de novo assemblies, including total assembly length, number of contigs (>100 bp), length of the largest contig, N50 (contig length at which 50% of the genome is covered), and number of ambiguous bases (# N's per 100 kbp). Table S4. Assembly metrics of Flye and Redbean meta-assemblies for *P. jefferyi*. Table S5. Raptor genome assemblies used in phylogenetic analysis. Table S6. Flagstats mapping quality statistics of the reads of each of the 35 Philippine eagle samples to the reference genome meta-assembly. Table S7. Samtools coverage of the individually aligned 35 Philippine eagle samples to the reference genome meta-assembly. % Bases covered refers to the overall percentage of the reference genome covered by reads. Figure S1. Graphical representations of Philippine eagle metadata. (A) Pedigree of sampled eagles showing sex (male, blue; female, pink) and three family units involving 15 individuals. The rest (20 individuals) are not directly related. (B) Map of sample sites across Mindanao and Samar Islands. Figure S2. BUSCO assessment metrics. Shown are proportions of complete (C: single-copy [S], duplicated [D]), fragmented (F), and missing (M) genes, relative to the total number of BUSCO groups searched (n) of the Flye meta-assembly, compared to the individual de novo assemblies. Figure S3. Comparison of genome-wide heterozygosity estimates of the 35 Philippine eagle samples across three computational methods. Genome-wide heterozygosity estimated using ANGSD with filters (Fold_Filt_Het; reported in main text), ANGSD without filters (Fold_Het), and the hard-call bcftools approach from [55] (Zou_Het). Individual samples are represented by points colored along a red-to-blue gradient by sample code, with outliers labeled. Lines connecting the same samples across methods demonstrate that the relative ranking of heterozygosity is preserved regardless of computational approach. The inset heatmap shows pairwise Pearson correlation coefficients between methods ($r > 0.99$ for all comparisons). Figure S4. Demographic history of Philippine eagle populations inferred from pairwise sequentially Markovian coalescent (PSMC) analysis. PSMC estimates of historical effective population size (N_e) over time for 35 Philippine eagles from two populations: Mindanao ($n=34$, red lines) and Samar ($n=1$, blue lines). Each line represents the demographic trajectory inferred from a single diploid genome. The x-axis shows time before present in years (log scale), and the y-axis shows effective population size in units of 10^4 individuals. PSMC analysis was performed using a mutation rate of 4.6×10^{-9} per base pair per generation and a generation time of 18 years.

Acknowledgements

We would like to thank the biologists and volunteer veterinarians at the Philippine Eagle Foundation Inc. for providing Philippine eagle samples, the University of the Philippines System—Philippine Genome Center for partial funding support, and MGI for providing DNBSEQ sequencing kits.

Authors' contributions

Conceptualization: DMP and CPS; sample collection: JL; methodology and formal analysis: KP, SMA, JL, MCA, RJM, JME, CL, DMP, and FP; writing—original draft: KP, SMA, JL, MCA, DMP, FP, CL, and FT; writing—review and editing, all authors; supervision: FT, JCG, JL, and CPS.

Funding

Open access funding provided by the University of the Philippines System. The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability

Draft genome sequence has been submitted to NCBI and should be accessible under BioProject PRJNA1348347.

Declarations

Ethics approval and consent to participate

The PEF has a 10-year Memorandum of Agreement (MOA) with the Philippine Department of Environment and Natural Resources (DENR), which provides the legal and ethical basis to undertake annual physical and medical screening for Philippine eagles as well as undertake genetic studies, species rehabilitation, captive-breeding, and species reintroduction (MOA No. SENR067470).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 11 November 2025 / Accepted: 9 April 2026

Published online: 20 April 2026

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